XIII. Methods to Evaluate Estimators (GSE 518)

Carlos Flores
Department of Economics
California Polytechnic State University*

1. Methods to Evaluate Estimators: Introduction

Setting: Assume $(X_1, \ldots, X_n) \overset{\text{iid}}{\sim} f(x \mid \theta)$. Say we have many estimators $\widehat{\theta}_{\text{MLE}}, \widehat{\theta}_{\text{MOM}}, \widehat{\theta}_A, \ldots$

- How do we evaluate estimators?
  - Which one is "better"?
  - Is there a "best" estimator?
  - In both cases, in what sense?
- Recall: $\widehat{\theta} = T(X_1, X_2, \ldots, X_n)$ is a random variable.
  - Ideally: We want the distribution of $\widehat{\theta}$ to be *centered* and *concentrated* around θ_0
  - Problem: θ_0 is unknown, so we do not know where it lies.

*MSQE Alumni Hunter Happersett contributed significantly to writing these lecture notes in Latex.

2. How to Compare Estimators

Definition: Let $\widehat{\theta}(X)$ be an estimator of θ_0. The *mean square error (MSE)* of $\widehat{\theta}$ is:

$$\text{MSE}(\widehat{\theta}) = E\left[(\widehat{\theta} - \theta_0)^2\right]$$

Notes on MSE:

1. It measures the average of the square of the error of $\widehat{\theta}$, where the error is $\widehat{\theta} - \theta_0$.

2. Since $\text{MSE}(\widehat{\theta})$ is an expectation, it is a function of the parameter θ_0.

3. The MSE can be written as:

$$\text{MSE}(\widehat{\theta}) = \text{Var}(\widehat{\theta}) + \left[E(\widehat{\theta}) - \theta_0\right]^2$$

Letting $Bias(\widehat{\theta}) = E(\widehat{\theta}) - \theta_0$, we can further write the MSE as:

$$\boxed{MSE(\widehat{\theta}) = \text{Var}(\widehat{\theta}) + Bias^2(\widehat{\theta})}$$

- Proof: To show:

  Recall that:

$$E(Y^2) = \text{Var}(Y) + [E(Y)]^2$$

  Let $Y = \widehat{\theta} - \theta_0$. Then,

$$E\left[(\widehat{\theta} - \theta_0)^2\right] = \text{Var}(\widehat{\theta} - \theta_0) + [E(\widehat{\theta} - \theta_0)]^2$$

$$\text{MSE}(\widehat{\theta}) = \text{Var}(\widehat{\theta}) + [E(\widehat{\theta}) - \theta_0]^2 = \text{Var}(\widehat{\theta}) + \text{Bias}^2(\widehat{\theta})$$

- *Intuition:*

$$MSE(\hat{\theta}) = Var(\hat{\theta}) + Bias^2(\hat{\theta})$$

"Precision" of $\hat{\theta}$ "Accuracy" of $\hat{\theta}$

Figure 1: MSE Intuition

  - "Precision" of $\widehat{\theta}$: How sparse is the distribution of $\widehat{\theta}$? How *concentrated* is the pdf of $\widehat{\theta}$?
  - "Accuracy" of $\widehat{\theta}$: Where is the pdf of $\widehat{\theta}$ *centered*?

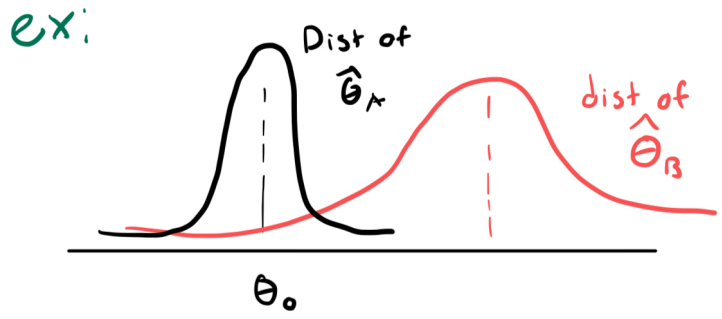


Figure 2: Distribution of Estimators

- Example:
  - $\widehat{\theta}_A$ has a high precision (low variance) and high accuracy (small bias).
  - $\widehat{\theta}_B$ has low precision (high variance) and low accuracy (large bias).

4. There are other measures of quality of estimators:

   - Mean Absolute Error (MAE) $= E[|\widehat{\theta} - \theta_0|]$
   - MSE is most used, however, since it is easy to work with and has a nice interpretation.

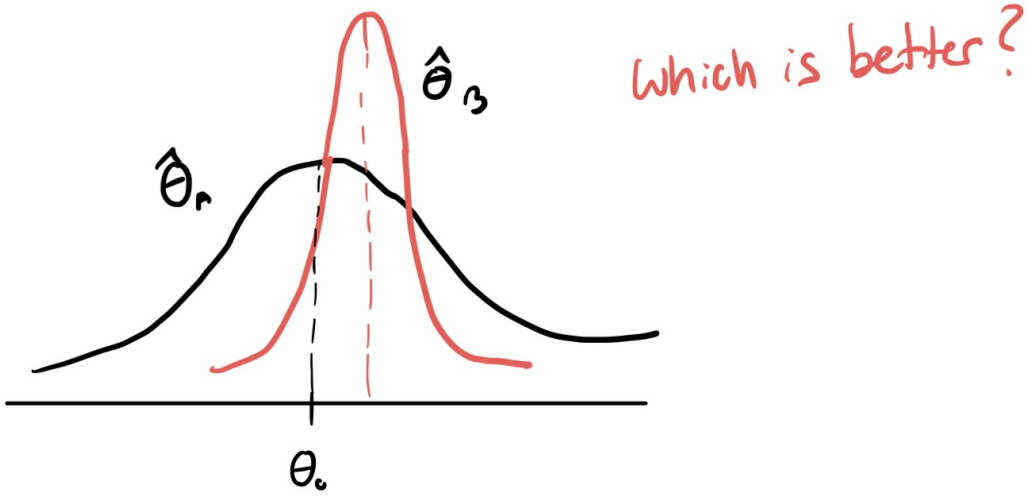
5. Definition of Bias $Bias(\widehat{\theta}) = E(\widehat{\theta}) - \theta_0$. If $E(\widehat{\theta}) = \theta_0$, then$= Bias(\widehat{\theta}) = 0$ and we say the estimator $\widehat{\theta}$ is *unbiased*.

   Notes on Bia:

   - It means the pdf of $\widehat{\theta}$ is centered at θ_0.
   - If the estimator is unbiased, then $MSE(\widehat{\theta}) = \text{Var}(\widehat{\theta})$

6. Usually there is a trade-off between bias and variance.

   - Example 1:



- Example 2: Let $(X_1, \ldots, X_n) \overset{\text{iid}}{\sim} N(\mu_0, \sigma_0^2)$. Two estimators of σ_0^2 we have seen are:

$$\widetilde{S}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2 \quad \text{and} \quad \widehat{S}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

Which one is better in terms of MSE?

(See solution from lecture)

- The bias-variance trade-off makes finding a "best" estimation very hard.

3. Best Unbiased Estimators

3.1. Motivation

- Ideally, we want an estimator $\widehat{\theta}^*$ of θ_0 with the *minimum MSE* as compared to *all possible estimators* $\widehat{\theta}$ *of* θ_0 and *for every possible value of* θ_0. That is,

$$MSE(\widehat{\theta}^*) \leq MSE(\widehat{\theta})) \qquad \text{for all } \widehat{\theta} \text{ and all } \theta_0 \in \Theta$$

  - Unfortunately, such estimator does not exist.
    - The class of estimators $\widehat{\theta}$ allowed is way too big.
    - In essence, we are asking way too much.

- Example: Let $(X_1, \ldots, X_n) \overset{iid}{\sim} Exponential(\mu_0)$, which implies $E(X) = \mu_0$ and $Var(X) = \mu_0^2$. Consider two estimators of μ_0:

  1. $\widehat{\mu}_1 = \overline{X}$
  2. $\widehat{\mu}_2 = 8$ (a "silly" estimator)

Which one is better in terms of MSE?

- From lecture: $MSE(\widehat{\mu}_1) = \mu_0^2/n$ and $MSE(\widehat{\mu}_2) = (8 - \mu_0)^2$
- Hence, which one is better depends on the values of θ_0. For example, if $\theta_0 = 8$, $\widehat{\mu}_2$ is better.
- Hence, we cannot say that $\widehat{\mu}_1$ is better than the silly estimator $\widehat{\mu}_2$ for *all* values of $\theta_0 \in \Theta$.
- This is what we mean by "the class of estimators allowed is too big".

- To rule out "silly" estimators and make progress towards finding a "best" estimator in a given class, we usually restrict ourselves to *unbiased* estimators of θ_0, and try to find the 'best among them.

3.2. Definition of Best Unbiased Estimator

Definition: The best unbiased estimator $\widehat{\theta}^*$ for $\theta_0 \in \Theta$ is one that:

a) $E(\widehat{\theta}^*) = \theta_0$ for all $\theta_0 \in \Theta$ (It's unbiased)

b) $\text{Var}(\widehat{\theta}^*) \leq \text{Var}(\widehat{\theta})$ for all $\theta_0 \in \Theta$ and any other *unbiased* estimator $\widehat{\theta}$.

$\widehat{\theta}^*$ is called *Uniform Minimum Variance Unbiased Estimator (UMVUE)*.

Notes on UMVUE:

- "Uniform" means for all $\theta_0 \in \Theta$.
- Here we only look at *unbiased* estimators.
  * Benefits: (i) We eliminate "silly" estimators. (ii) In many cases, we can find a UMVUE.
  - Cost[*]: We may rule out some good estimators (e.g., $\widetilde{S}^2$ when estimating σ_0^2).

- Unbiasedness $\implies MSE(\widehat{\theta}) = \text{Var}(\widehat{\theta})$

3.3. How to Find a UMVUE ($\widehat{\theta}^*$)?

Cramer-Rao Inequality (iid case).

Let $(X_1, \ldots, X_n) \overset{iid}{\sim} f(x \mid \theta_0)$, $\theta_0 \in \Theta$. Let $\widehat{\theta}$ be any unbiased estimator of θ_0. Then, under some conditions:

$$\text{Var}(\widehat{\theta}) \geq \frac{1}{nE[S^2(\theta_0, x)]}$$

The term $\frac{1}{nE[S^2(\theta_0, x)]}$ is called the *Cramer-Rao Lower Bound (CRLB)*

Notes on CRLB:

1. By score identity, $E[S(\theta_0, X)] = 0$, then

$$E[S^2(\theta_0, X)] = Var[S(\theta_0, X)]$$

2. By the information matrix identity,

$$\frac{1}{nE[S^2(\theta_0, X)]} = \frac{1}{-nE\left[\frac{\partial S(\theta, X)}{\partial \theta}\right]}$$

3. The quantity $E[S^2(\theta, X)]$ is called *"Fisher's Information Matrix for observation i"*, and it is usually written as $\mathcal{I}(\theta_0)$.

   Note that we can calculate $\mathcal{I}(\theta_0)$ as (note all objects below are evaluated at θ_0):

   a) $E[S^2(\theta_0, X)]$

   b) $Var[S(\theta_0, X)]$

   c) $-E\left[\frac{\partial S(\theta_0, X)}{\partial \theta}\right]$

4. Key: How do we use the CRLB to find a UMVUE? The CRLB gives a *lower bound* on the variance of *any* unbiased estimator. Therefore, if we find an estimator $\widehat{\theta}$ such that:

   1) $E[\widehat{\theta}] = \theta_0$ ($\widehat{\theta}$ is unbiased)

   2) $Var(\widehat{\theta}) = \dfrac{1}{n\mathcal{I}(\theta_0)}$ ($Var(\widehat{\theta}) = CRLB$, i.e., $Var(\widehat{\theta})$ achieves the lower bound)

   Then, $\widehat{\theta}$ is UMVUE.

   This means that there is no other unbiased estimator that has lower variance than the UMVUE.

5. The converse of the point above is not true. If $\widehat{\theta}^*$ is the UMVUE $\not\Rightarrow Var(\widehat{\theta}^*) = \frac{1}{n\mathcal{I}(\theta_0)}$.

6. The theorem assumes that we can interchange the order of derivative and integral. Hence, for example, if the parameter space of X is a function of θ_0, the theorem does not apply.

$$X \sim \text{Uniform}(0, \theta_0)$$

$$f(x) = \begin{cases} \frac{1}{\theta_0} & \text{if } x \in (0, \theta_0) \\ 0 & \text{otherwise} \end{cases}$$

7. To summarize: If $(X_1, \ldots, X_n) \overset{iid}{\sim} f(x \mid \theta_0)$ (i.e., in the iid case):

   i) CRLB $= \frac{1}{n\mathcal{I}(\theta_0)}$, where we can calculate $\mathcal{I}(\theta_0)$ in three possible ways:

   $\mathcal{I}(\theta_0) = E[S^2(\theta_0, X)]$

   $\mathcal{I}(\theta_0) = \text{Var}[S(\theta_0, X)]$

$$\mathcal{I}(\theta_0) = -E\left[\frac{\partial S(\theta_0, X)}{\partial \theta}\right]$$

ii) A useful way to find a UMVUE:

If $\widehat{\theta}$ is unbiased and $Var(\widehat{\theta}) = \text{CRLB} \implies \widehat{\theta}$ is the UMVUE

8. Example: Let $(X_1, \ldots, X_n) \overset{iid}{\sim} N(\mu_0, \sigma^2)$, where we assume that σ^2 is *known*.

a) Find the CRLB for any unbiased estimator $\widehat{\mu}$ of μ_0.

Steps (*See details in lecture*)

(a) Get Socre function

(b) Get Fisher's Information matrix (you may use any of the 3 ways available to calculate it)

(c) Get the CRLB

b) Is $\overline{X}$ a UMVUE of μ_0?